

Traian Florin Șerbănuță

List of Publications

traian.serbanuta@unibuc.ro · <http://cs.unibuc.ro/~tserbanuta> · github.com/traiansf

Publications

55 articles · h-index 22 · 2178 citations (Google Scholar, as of October 2026)

Reverse chronological within each category. Citation counts are from Google Scholar (October 2026); see also DBLP.

Journal Articles

1. Leuştean, Ioana, Natalia Moangă, and Traian Florin Șerbănuță. “Many-sorted hybrid modal languages.” *Journal of Logical and Algebraic Methods in Programming*, Vol. 120, 100644, 2021. [3]
2. Leuştean, Ioana, Natalia Moangă, and Traian Florin Șerbănuță. “A many-sorted polyadic modal logic.” *Fundamenta Informaticae*, Vol. 173, No. 2-3, pp. 191-215, 2020. [12]
3. Ștefănescu, Andrei, Ștefan Ciobâcă, Radu Mereuta, Brandon M Moore, Traian Florin Șerbănuță, and Grigore Roșu. “All-Path Reachability Logic.” *Logical Methods in Computer Science*, Vol. 15, Issue 2, 2019. [117]
4. Arusoiaie, Andrei, Ștefan Ciobâcă, Dorel Lucanu, Grigore Roșu, Vlad Rusu, and Traian-Florin Șerbănuță. “Program Logics and Their Applications.” *Revue Roumaine de Mathématiques Pures et Appliquées*, Vol. 62, No. 1, pp. 137-154, 2017.
5. Rusu, Vlad, Dorel Lucanu, Traian-Florin Șerbănuță, Andrei Arusoiaie, Andrei Ștefănescu, and Grigore Roșu. “Language Definitions as Rewrite Theories.” *Journal of Logical and Algebraic Methods in Programming*, Vol. 85, No. 1, pp. 98-120, 2016. [34]
6. Frei, Regina, Traian Florin Șerbănuță, and Giovanna Di Marzo Serugendo. “Self-organising assembly systems formally specified in Maude.” *Journal of Ambient Intelligence and Humanized Computing*, Vol. 5, No. 4, pp. 491-510, 2014. [12]
7. Frei, Regina, Giovanna Di Marzo Serugendo, and Traian Florin Șerbănuță. “Ambient intelligence in self-organising assembly systems using the chemical reaction model.” *Journal of Ambient Intelligence and Humanized Computing*, Vol. 1, No. 3, pp. 163-184, 2010. [26]
8. Roșu, Grigore and Traian Florin Șerbănuță. “An Overview of the K Semantic Framework.” *Journal of Logic and Algebraic Programming*, Vol. 79, No. 6, pp. 397-434, 2010. [673]
9. Chira, Camelia, Traian Florin Șerbănuță, and Gheorghe Ștefănescu. “P systems with control nuclei: The concept.” *Journal of Logic and Algebraic Programming*, Vol. 79, No. 6, pp. 326-333, 2010. [9]
10. Popescu, Andrei, Traian Florin Șerbănuță, and Grigore Roșu. “A Semantic Approach to Interpolation.” *Theoretical Computer Science*, Vol. 410, No. 12-13, pp. 1109-1128, 2009. [20]
11. Șerbănuță, Traian Florin, Grigore Roșu, and José Meseguer. “A Rewriting Logic Approach to Operational Semantics.” *Information and Computation*, Vol. 207, No. 2, pp. 305-340, 2009. [129]
12. Șerbănuță, Virgil Nicolae and Traian Florin Șerbănuță. “Injectivity of the Parikh matrix mappings revisited.” *Fundamenta Informaticae*, Vol. 73, No. 1-2, pp. 265-283, 2006. [63]
13. Șerbănuță, Traian Florin. “Extending Parikh matrices.” *Theoretical Computer Science*, Vol. 310, No. 1-3, pp. 233-246, 2004. [66]

Conference and Workshop Papers

1. Tușil, Jan, Traian Florin Șerbănuță, and Jan Obdržálek. “Cartesian Reachability Logic: A Language-parametric Logic for Verifying k-Safety Properties.” *Logic for Programming, Artificial Intelligence and Reasoning (LPAR 2023)*, pp. 405-456, 2023. [1]

2. Trufaş, Dafina, Ioan Teodorescu, Denisa Diaconescu, Traian Florin Şerbănuţă, and Vlad Zamfir. “Asynchronous Muddy Children Puzzle (work in progress).” *Working Formal Methods Symposium (FROM 2023)*, EPTCS Vol. 389, pp. 152-166, 2023.
3. Li, Elaine, Traian Florin Şerbănuţă, Denisa Diaconescu, Vlad Zamfir, and Grigore Roşu. “Formalizing Correct-by-Construction Casper in Coq.” *IEEE International Conference on Blockchain and Cryptocurrency (ICBC 2020)*, pp. 1-3, 2020. **[19]**
4. Leuştean, Ioana, Natalia Moangă, and Traian Florin Şerbănuţă. “From Hybrid Modal Logic to Matching Logic and Back.” *Working Formal Methods Symposium (FROM 2019)*, EPTCS Vol. 303, pp. 16-31, 2019. **[3]**
5. Leuştean, Ioana, Natalia Moangă, and Traian Florin Şerbănuţă. “Operational Semantics and Program Verification Using Many-Sorted Hybrid Modal Logic.” *Automated Reasoning with Analytic Tableaux and Related Methods (TABLEAUX’19)*, 2019. **[10]**
6. Kasampalis, Theodoros, Dwight Guth, Brandon Moore, Traian Florin Şerbănuţă, Yi Zhang, Daniele Filaretti, Virgil Şerbănuţă, Ralph Johnson, and Grigore Roşu. “IELE: A Rigorously Designed Language and Tool Ecosystem for the Blockchain.” *Formal Methods (FM’19)*, pp. 593-610, 2019. **[46]**
7. Roşu, Grigore and Traian Florin Şerbănuţă. “Matching Logic: Syntax and Semantics.” *Working Formal Methods Symposium (FROM 2017)*, 2017.
8. Daian, Philip, Dwight Guth, Chris Hathhorn, Yilong Li, Edgar Pek, Manasvi Saxena, Traian Florin Şerbănuţă, and Grigore Roşu. “Runtime Verification at Work: A Tutorial.” *Runtime Verification (RV’16)*, pp. 46-67, 2016. **[12]**
9. Şerbănuţă, Traian Florin and Liviu P. Dinu. “Maximally Parallel Contextual String Rewriting.” *Rewriting Logic and Its Applications (WRLA’16)*, pp. 152-166, 2016.
10. Daian, Philip, Ylies Falcone, Patrick Meredith, Traian Florin Şerbănuţă, Akihito Iwai, Shin’ichi Shiriashi, and Grigore Roşu. “RV-Android: Efficient Parametric Android Runtime Verification, a Brief Tutorial.” *Runtime Verification (RV’15)*, LNCS Vol. 9333, pp. 342-357, 2015. **[31]**
11. Chiriţă, Claudia Elena and Traian Florin Şerbănuţă. “An Institutional Foundation for the K Semantic Framework.” *Recent Trends in Algebraic Development Techniques (WADT’14)*, LNCS Vol. 9463, pp. 9-29, 2015. **[4]**
12. Luo, Qingzhou, Yi Zhang, Choonghwan Lee, Dongyun Jin, Patrick O’Neil Meredith, Traian Florin Şerbănuţă, and Grigore Roşu. “RV-Monitor: Efficient Parametric Runtime Verification with Simultaneous Properties.” *Runtime Verification (RV’14)*, LNCS Vol. 8734, pp. 285-300, 2014. **[108]**
13. Ştefănescu, Andrei, Ştefan Ciobâcă, Radu Mereuta, Brandon M Moore, Traian Florin Şerbănuţă, and Grigore Roşu. “All-Path Reachability Logic.” *Rewriting and Typed Lambda Calculi (RTA-TLCA’14)*, LNCS Vol. 8560, pp. 425-440, 2014.
14. Arusoai, Andrei, Dorel Lucanu, Vlad Rusu, Traian-Florin Şerbănuţă, Andrei Ştefănescu, and Grigore Roşu. “Language Definitions as Rewrite Theories.” *Rewriting Logic and Its Applications (WRLA’14)*, LNCS, pp. 97-112, 2014.
15. Roşu, Grigore and Traian Florin Şerbănuţă. “K Overview and SIMPLE Case Study.” *Proceedings of K’11*, ENTCS Vol. 304, pp. 3-56, 2014. **[64]**
16. Şerbănuţă, Traian Florin, Andrei Arusoai, David Lazar, Chucky Ellison, Dorel Lucanu, and Grigore Roşu. “The K Primer (version 3.3).” *Proceedings of K’11*, ENTCS Vol. 304, pp. 57-80, 2014. **[54]**
17. Şerbănuţă, Traian Florin. “Rewriting Semantics and Analysis of Concurrency Features for a C-like Language.” *Proceedings of K’11*, ENTCS Vol. 304, pp. 167-182, 2014. **[2]**
18. Şerbănuţă, Traian Florin, Feng Chen, and Grigore Roşu. “Maximal Causal Models for Sequentially Consistent Systems.” *Runtime Verification (RV’12)*, LNCS Vol. 7687, pp. 136-150, 2013. **[105]**
19. Şerbănuţă, Traian Florin. “Programming Language Semantics using K — true concurrency through term graph rewriting.” EPTCS Vol. 110, p. 2, 2013.
20. Şerbănuţă, Traian Florin and Grigore Roşu. “A Truly Concurrent Semantics for the K Framework Based on Graph Transformations.” *Graph Transformations (ICGT’12)*, LNCS Vol. 7562, pp. 294-310, 2012. **[16]**

21. Lazar, David, Andrei Arusoaie, Traian Florin Șerbănuță, Chucky Ellison, Radu Mereuta, Dorel Lucanu, and Grigore Roșu. "Executing Formal Semantics with the K Tool." *Formal Methods (FM'12)*, LNCS Vol. 7436, pp. 267-271, 2012. [21]
22. Lucanu, Dorel, Traian Florin Șerbănuță, and Grigore Roșu. "K Framework Distilled." *Rewriting Logic and Its Applications (WRLA'12)*, LNCS Vol. 7571, pp. 31-53, 2012. [29]
23. Arusoaie, Andrei, Traian Florin Șerbănuță, Chucky Ellison, and Grigore Roșu. "Making Maude Definitions More Interactive." *Rewriting Logic and Its Applications (WRLA'12)*, LNCS Vol. 7571, pp. 83-98, 2012. [4]
24. Șerbănuță, Traian Florin and Grigore Roșu. "K-Maude: A Rewriting Based Tool for Semantics of Programming Languages." *Rewriting Logic and Its Applications (WRLA'10)*, LNCS Vol. 6381, pp. 104-122, 2010. [74]
25. Ellison, Chucky, Traian Florin Șerbănuță, and Grigore Roșu. "A Rewriting Logic Approach to Type Inference." *Recent Trends in Algebraic Development Techniques (WADT'08)*, LNCS Vol. 5486, pp. 135-151, 2009. [28]
26. Roșu, Grigore, Wolfram Schulte, and Traian Florin Șerbănuță. "Runtime Verification of C Memory Safety." *Runtime Verification (RV'09)*, LNCS Vol. 5779, pp. 132-151, 2009. [64]
27. Șerbănuță, Traian Florin, Gheorghe Ștefănescu, and Grigore Roșu. "Defining and Executing P Systems with Structured Data in K." *Membrane Computing (WMC'08)*, LNCS Vol. 5391, pp. 374-393, 2009. [28]
28. Chen, Feng, Traian Florin Șerbănuță, and Grigore Roșu. "jPredictor: a predictive runtime analysis tool for Java." *ICSE '08: Proceedings of the 30th International Conference on Software Engineering*, pp. 221-230, 2008. [155]
29. Ghoshal, Sudipto, Solaiappan Manimaran, Grigore Roșu, Traian Florin Șerbănuță, and Gheorghe Ștefănescu. "Monitoring IVHM Systems using a Monitor-Oriented Programming Framework." *Sixth NASA Langley Formal Methods Workshop (LFM 2008)*, 2008. [5]
30. Șerbănuță, Traian Florin, Grigore Roșu, and José Meseguer. "A Rewriting Logic Approach to Operational Semantics (Extended Abstract)." *Structural Operational Semantics (SOS'07)*, ENTCS Vol. 192, No. 1, pp. 125-141, 2007.
31. Hills, Mark, Traian Florin Șerbănuță, and Grigore Roșu. "A Rewrite Framework for Language Definitions and for Generation of Efficient Interpreters." *Rewriting Logic and Its Applications (WRLA'06)*, ENTCS Vol. 176, No. 4, pp. 215-231, 2007. [31]
32. Denker, Grit, et al. "Rewriting Logic Systems." *Rewriting Logic and Its Applications (WRLA'06)*, ENTCS Vol. 176, No. 4, 2007. [7]
33. Șerbănuță, Traian Florin and Grigore Roșu. "Computationally Equivalent Elimination of Conditions." *Rewriting Techniques and Applications (RTA'06)*, LNCS Vol. 4098, pp. 19-34, 2006. [41]
34. Popescu, Andrei, Traian Florin Șerbănuță, and Grigore Roșu. "A Semantic Approach to Interpolation." *Foundations of Software Science and Computation Structures (FoSSaCS'06)*, LNCS Vol. 3921, pp. 307-321, 2006.
35. Reitter, David, Ștefan Covaci, Florin Oltean, Cătălin Băcanu, and Traian Florin Șerbănuță. "Hybrid Natural Language Processing in a Customer-Care Environment." *11th Student Conference on Computational Linguistics (TaCoS'01)*, 2001. [7]

Books, Book Chapters, and Theses

1. Lucanu, Dorel, Traian-Florin Șerbănuță, and Grigore Roșu. "Towards a Kool Future." In *Theory and Practice of Formal Methods*, LNCS, Springer, pp. 325-343, 2016. [1]
2. Șerbănuță, Traian Florin. *Programming Language Design and Analysis: A Rewriting Approach*. Editura Universității „Alexandru Ioan Cuza”, Iași, 2012. Revised edition of the PhD thesis.
3. Șerbănuță, Traian Florin. *A Rewriting Approach to Concurrent Programming Language Design and Semantics*. PhD thesis, University of Illinois at Urbana-Champaign, 2010. [24]
4. Șerbănuță, Traian Florin. *Institutional Concepts in First-Order Logic, Parameterized Specifications, and Logic Programming*. Master's dissertation, University of Bucharest, 2004. [5]

5. Șerbănuță, Traian Florin. *Ascunderea informației în text folosind gramatici de tip LR(k)* (Information Hiding in Text Using LR(k) Grammars). BSc thesis, University of Bucharest, 2002.

Preprints

1. Șerbănuță, Traian Florin, Jun Xu, Andrei Ștefănescu, and Cosmin Radoi. “Solving VeriContest with a Lean-Backed Rust Verifier.” arXiv:2610.03994, 2026.
2. Zamfir, Vlad, Mihai Calancea, Denisa Diaconescu, Wojciech Kołowski, Brandon Moore, Karl Palmskog, Traian Florin Șerbănuță, Michael Stay, Dafina Truș, and Jan Tušil. “Validating Labelled State Transition and Message Production Systems: A Theory for Modelling Faulty Distributed Systems.” arXiv:2202.12662, 2022. [1]

Technical Reports

1. Arusoaie, Andrei and Traian Florin Șerbănuță. “Contextual Transformation in K Framework.” 2011. [3]
2. Șerbănuță, Traian Florin, Feng Chen, and Grigore Roșu. “Maximal Causal Models for Sequentially Consistent Systems.” University of Illinois, 2011.
3. Șerbănuță, Traian Florin, Feng Chen, and Grigore Roșu. “Maximal Causal Models for Sequentially Consistent Multithreaded Systems.” University of Illinois, 2010.
4. Șerbănuță, Traian Florin and Grigore Roșu. “KRAM — Extended Report.” University of Illinois, 2010.
5. Roșu, Grigore, Wolfram Schulte, and Traian Florin Șerbănuță. “Runtime Verification of C Memory Safety.” UIUCDCS-R-2009-3048, University of Illinois, 2009.
6. Șerbănuță, Traian Florin, Feng Chen, and Grigore Roșu. “Maximal Causal Models for Multithreaded Systems.” UIUCDCS-R-2008-3017, University of Illinois, 2008.
7. Ellison, Chucky, Traian Florin Șerbănuță, and Grigore Roșu. “A Rewriting Logic Approach to Type Inference.” UIUCDCS-R-2008-2934, University of Illinois, 2008.
8. Chen, Feng, Traian Florin Șerbănuță, and Grigore Roșu. “Effective Predictive Runtime Analysis Using Sliced Causality and Atomicity.” UIUCDCS-R-2007-2905, University of Illinois, 2007. [3]
9. Șerbănuță, Traian Florin and Grigore Roșu. “A Rewriting Logic Approach to Operational Semantics.” UIUCDCS-R-2006-2780, University of Illinois, 2006.
10. Șerbănuță, Traian Florin and Grigore Roșu. “Computationally Equivalent Elimination of Conditions.” UIUCDCS-R-2006-2693, University of Illinois, 2006.
11. Popescu, Andrei, Traian Florin Șerbănuță, and Grigore Roșu. “A Semantic Approach to Interpolation.” UIUCDCS-R-2005-2643, University of Illinois, 2005.
12. Șerbănuță, Traian Florin. “Report on an Envelope Invariance Checker Tool for Specifications in PMEL.” 2005.
13. Șerbănuță, Traian Florin and Grigore Roșu. “Towards Effectively Eliminating Conditional Rewrite Rules.” UIUCDCS-R-2004-2494, University of Illinois, 2004.